Beyond TRIAGE-FHIR: Supervised Fine-Tuning of Adaptive Agents for FHIR Clinical Question Answering

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1 Introduction

Electronic Health Records (EHRs) increasingly adopt the HL7 Fast Healthcare Interoperability Resources (FHIR) standard [1], whose complicated, interconnected resource structure poses challenges for reliable clinical reasoning for large language models (LLMs). Our previous CPSC 5770 final project, TRIAGE-FHIR, addressed this challenge through a tri-agent architecture that separates querying, verification, and answer generation, improving accuracy while providing explainable intermediate reasoning. However, the orchestration of the tri-agent framework is manually designed, resulting in a largely predetermined computation path across clinical questions with varying reasoning demands. In this project, we move beyond predefined clinical agent workflows toward adaptive reasoning systems that dynamically adjust their reasoning and tool use based on each specific query. We develop a trainable tool-use agent based on the open-weight Qwen3-8B model and investigate trajectory-level supervised fine-tuning (SFT) as a lightweight post-training approach for learning such adaptive behavior.

2 Related work

Recent benchmarks such as FHIR-AgentBench [2] and MedAgentBench [3] show that LLM agents remain unreliable for FHIR-based EHR reasoning, with persistent challenges in resource identification, reference traversal, and reasoning. Prior work has addressed these limitations through both modular architecture design and post-training. EHRNavigator introduces an end-to-end multi-agent framework that decomposes complex tasks into modular sub-tasks [4], while Knorr et al. implements a CodeAct agent and post-trains it with reinforcement learning (RL), achieving 77% accuracy on FHIR-AgentBench [5]. However, RL depends on careful reward design and expensive online rollouts, and existing approaches primarily focus on answer accuracy. We therefore investigate whether a lightweight trajectory-level SFT can teach an agent to adjust its tool-use effort adaptively across questions of varying complexity.

3 Proposed approach

We propose to extend TRIAGE-FHIR from a rigid tri-agent workflow into a trainable, adaptive FHIR reasoning system. Specifically, we will:

1. **Reformulate TRIAGE-FHIR as a ReAct-style agent:** We will replace the predefined TRIAGE-FHIR workflow with a multi-turn interaction loop in which an open-weight Qwen3-8B model iteratively reasons and acts through tool calls. Based on intermediate observations, the agent decides whether to retrieve FHIR resources, perform computation, verify evidence, or produce the final answer.
2. **Post-train the agent with trajectory-level SFT:** We will collect successful interaction trajectories on the training set, retaining the shortest correct trajectory for each question so that the training data reflects computation matched to reasoning demands. Fine-tuning Qwen3-8B on these trajectories offers a lightweight alternative to RL without reward design or online rollouts.

3. **Evaluate adaptive reasoning and efficiency:** We will compare a predefined workflow with zero-shot and SFT-trained adaptive agents in terms of answer correctness and inference efficiency.

4 Experimental plan

We design our experiments to evaluate whether adaptive orchestration and trajectory-level SFT improve FHIR clinical question answering while enabling more flexible and efficient tool use across queries with varying reasoning demands.

1. **Data and Trajectory Construction:** We will extend our existing Synthea-based FHIR benchmark, which provides ground-truth answers and supporting FHIR resources for each question. The dataset will be divided into training, validation, and test sets. Successful multi-step interaction trajectories from the training set will be collected for supervised fine-tuning.
2. **Baselines and Comparisons:** Using the same Qwen3-8B backbone and FHIR environment, we will compare: (i) **Predefined Workflow**, which follows the manually designed TRIAGE-FHIR orchestration; (ii) **Zero-shot Adaptive Agent**, which dynamically determines its reasoning and tool-use strategy without post-training; and (iii) **SFT Adaptive Agent**, which is fine-tuned on successful interaction trajectories. These comparisons isolate the effects of adaptive orchestration and trajectory-level SFT.
3. **Evaluation:** We will evaluate **answer correctness** as the primary metric and **inference efficiency** using the number of tool interactions and generated tokens per question.

References

[1] "FHIR v5.0.0." Accessed: Mar. 06, 2026. [Online]. Available: https://www.hl7.org/fhir/

[2] G. Lee *et al.*, "FHIR-AgentBench: Benchmarking LLM Agents for Realistic Interoperable EHR Question Answering," Nov. 13, 2025, *arXiv*: arXiv:2509.19319. doi: 10.48550/arXiv.2509.19319.

[3] Y. Jiang *et al.*, "MedAgentBench: A Realistic Virtual EHR Environment to Benchmark Medical LLM Agents," Feb. 12, 2025, *arXiv*: arXiv:2501.14654. doi: 10.48550/arXiv.2501.14654.

[4] L. Qian *et al.*, "EHRNavigator: A Multi-Agent System for Patient-Level Clinical Question Answering over Heterogeneous Electronic Health Records," Jan. 15, 2026, *arXiv*: arXiv:2601.10020. doi: 10.48550/arXiv.2601.10020.

[5] M. S. Knorr *et al.*, "Reinforcement Learning for Tool-Calling Agents in Fast Healthcare Interoperability Resources (FHIR)," May 13, 2026, *arXiv*: arXiv:2605.14126. doi: 10.48550/arXiv.2605.14126.

AI Usage Statement

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